

A Proactive Multimodal AI Companion for Glycemic Management and Fear of Hypoglycemia Support

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Abstract

Fear of hypoglycemia can shape meals, physical activity and sleep, even when glucose readings appear stable. GlucoMind considers physiological risk and experienced fear through a **dynamic multimodal patient twin**, **coordinated agent reasoning** and **proactive support**. Our Android prototype connects wearable histories, editable health profiles, conversational memory and audited check-ins. An offline prediction research pipeline explores future glucose trajectories from recent history and contextual signals.

System Architecture

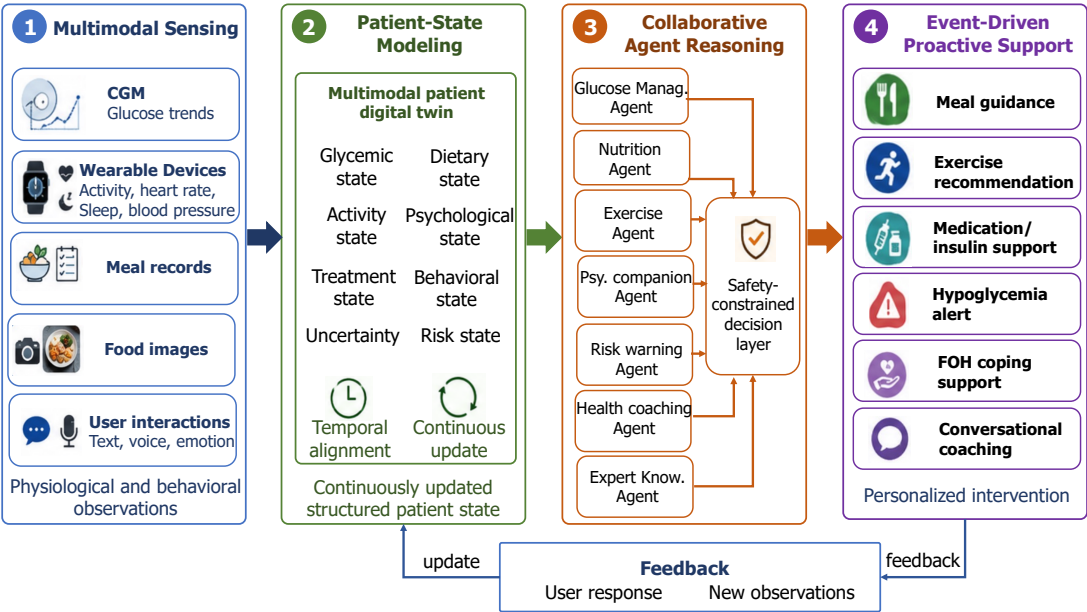
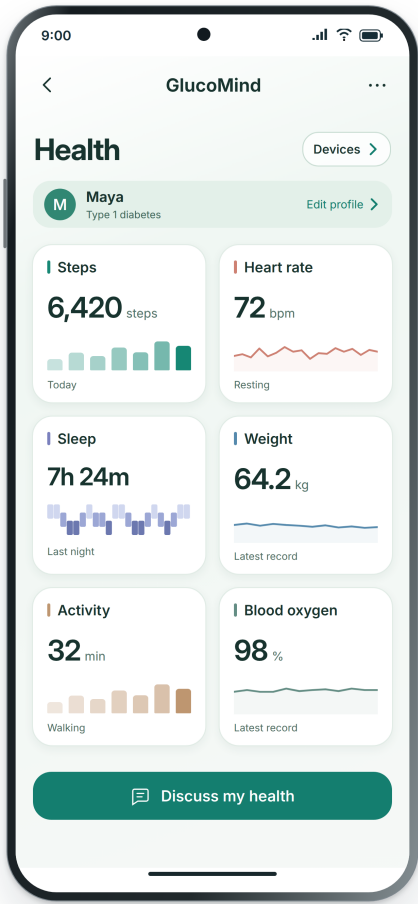


Figure 1: System architecture of GlucoMind

Prototype Features



Multimodal conversation

Photo and PDF report input in one continuous chat.

Profile and memory

Editable health facts and relevant conversational recall.

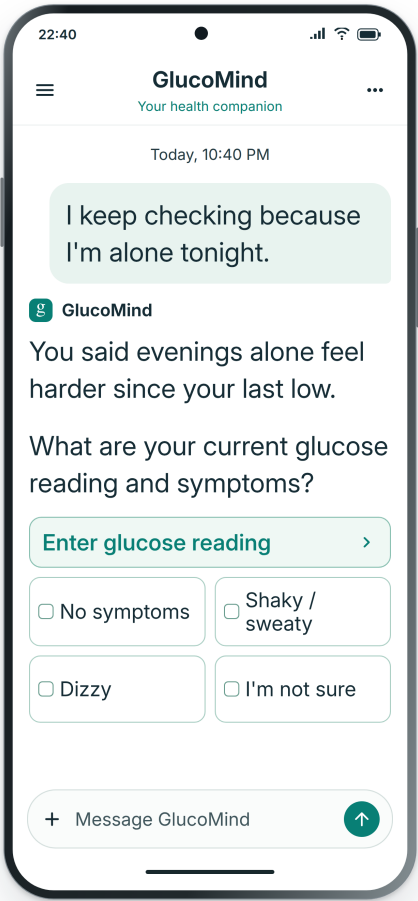
Wearable synchronization

10 data categories through Zepp and Health Connect.

Audited proactive care

Persistent follow-up tasks, timing controls and push delivery.

Application Scenarios



01 Dinner

“Should I eat more, just in case?”
Connect the question to a previous low.

02 Bedtime

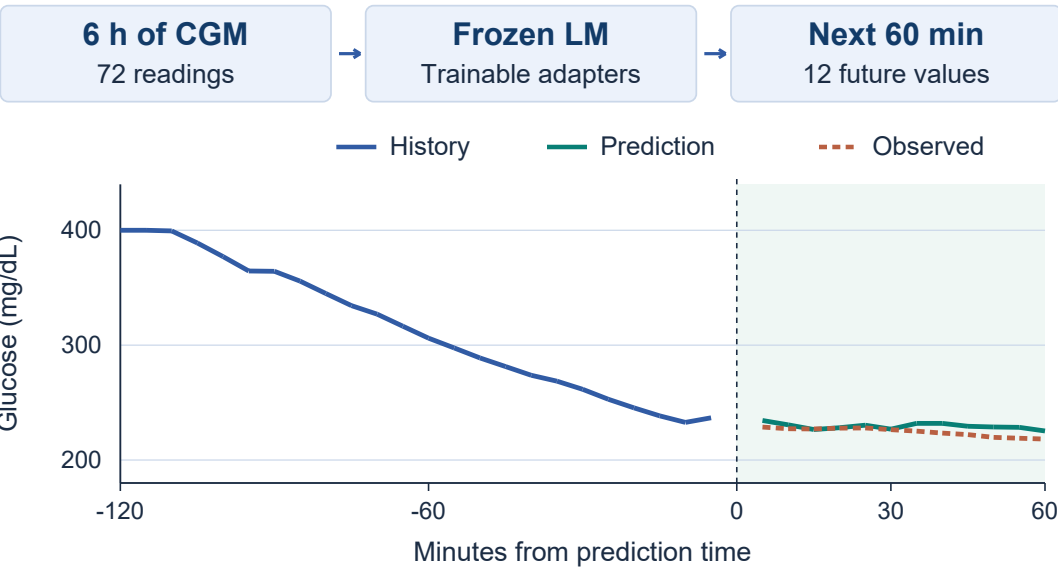
“I keep checking. I am alone tonight.”
Clarify current readings and acknowledge the concern.

03 Next morning

“How did last night feel?”
Continue the conversation at the agreed time.

Glucose Prediction

Offline Glucose prediction pipeline



Glucose prediction results on CGMacros dataset.

Last 2 h of history shown; prediction horizon: 5-60 min.

Contributions

- 1 Dynamic patient twin**
Physiological context linked to personal experience.
- 2 Coordinated agent reasoning**
Shared evidence, uncertainty and safety priorities.
- 3 Proactive, psychologically informed support**
Timely follow-up shaped by the person’s needs.

Future Work

- Live integration.** Connect CGM and prediction to patient-state updates and proactive support.
- Safety and usability.** Evaluate with clinicians and people living with diabetes.
- Longitudinal outcomes.** Study glycemic, psychological and engagement outcomes over time.